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Dose-response of iron supplementation on fatigue (corrected version)

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Abstract

Background: Fatigue is the most frequent complaint of adults with iron deficiency without anaemia, yet the amount of oral iron needed to relieve it is not settled. Most trials compare a single dose against placebo, which says little about how much iron is enough. Methods: We conducted a 12-week, double-blind, randomised trial in 150 iron-deficient, non-anaemic adults aged 18 to 55 years, allocated in equal numbers to 20, 40 or 60 mg of elemental iron daily. The primary outcome was the change in a validated fatigue score from baseline to week 12, modelled as a linear function of the daily dose. Secondary outcomes were serum ferritin, transferrin saturation, haemoglobin, adherence and gastrointestinal adverse events. Results: In 150 iron-deficient adults, fatigue scores fell with increasing iron dose. Ferritin rose in every arm and most in the highest-dose arm, while gastrointestinal complaints were more frequent at 60 mg. The estimated dose-response slope and its confidence interval are reported in the Results. Conclusions: Within the range studied, larger daily doses of oral iron produced larger reductions in fatigue, at the cost of more gastrointestinal complaints. Clinicians may weigh these two effects when choosing a dose.

1. Introduction

Iron deficiency is the most common nutritional deficiency worldwide and affects a large share of otherwise healthy adults, in particular women of reproductive age, frequent blood donors, vegetarians and people who train for endurance sports. In its early stage, iron stores are depleted while haemoglobin remains within the normal range. This state, iron deficiency without anaemia, is often missed because routine blood counts look normal, and its symptoms are non-specific.

Fatigue is the symptom that brings most of these patients to a clinician. Iron is needed not only for haemoglobin but for myoglobin, for the cytochromes of the respiratory chain and for several enzymes of neurotransmitter synthesis, so a shortage could plausibly cause tiredness before any anaemia appears. Several randomised trials have found that iron given to non-anaemic women with low ferritin reduces self-reported fatigue compared with placebo, and systematic reviews have reached similar conclusions, while noting that the trials were small and used different doses, formulations and fatigue scales.

What these trials cannot answer is how the benefit depends on dose. Clinical guidelines recommend daily doses that vary several-fold, from about 20 mg to more than 100 mg of elemental iron, and higher doses are known to cause more nausea, constipation and abdominal pain, which in turn lower adherence. Studies of iron absorption suggest that a large single dose raises circulating hepcidin and reduces the fraction of iron absorbed from the next dose, so the extra iron in a high dose may be partly wasted. If the effect on fatigue levels off at modest doses, patients could be spared the side effects of high doses with no loss of benefit.

We therefore designed a trial that randomised iron-deficient, non-anaemic adults to three daily doses of the same oral iron preparation and estimated the change in fatigue per unit of dose. Our primary hypothesis was that the reduction in fatigue would increase with dose across the range studied. Our secondary aims were to describe the corresponding changes in iron status and the frequency of adverse events at each dose.

2. Methods

2.1 Study design and setting

The trial was a single-centre, parallel-group, double-blind, randomised dose-comparison trial carried out at the outpatient nutrition clinic of Fake University Hospital. The protocol was approved by the institutional ethics committee and registered before the first participant was enrolled. All participants

gave written informed consent. The trial followed the principles of the Declaration of Helsinki and is reported according to the CONSORT statement for randomised trials.

2.2 Participants

Adults aged 18 to 55 years who reported fatigue for at least four weeks were screened. Eligible participants had a serum ferritin below 30 µg/L, a haemoglobin within the sex-specific reference range and a baseline fatigue score of at least 10 on the scale described below. We excluded people with a known cause of fatigue other than iron deficiency, including thyroid disease, depression under treatment, sleep apnoea and chronic inflammatory disease; people with a C-reactive protein above 10 mg/L; pregnant or breastfeeding women; and anyone who had taken iron supplements or received an iron infusion in the previous three months.

2.3 Randomisation and blinding

Participants were randomised in a 1:1:1 ratio to 20, 40 or 60 mg of elemental iron daily, using computer-generated permuted blocks of six stratified by sex. The allocation list was prepared by a statistician not involved in recruitment and held by the hospital pharmacy. All capsules were identical in size, colour and weight, and each participant took three capsules each morning, containing iron sulphate or an inert filler in the proportions needed for the allocated dose. Participants, clinic staff and outcome assessors were unaware of the allocation until the database was locked.

2.4 Intervention

The capsules were taken on an empty stomach with water for 12 weeks. Participants were advised to avoid tea, coffee and calcium-rich foods for one hour after the dose. Adherence was assessed by capsule counts at weeks 4, 8 and 12 and expressed as the percentage of the prescribed capsules taken.

Table 1. Allocated daily doses and capsule composition

Arm	Elemental iron per day	Iron capsules	Filler capsules	Participants randomised
Low dose	20 mg	1	2	50
Middle dose	40 mg	2	1	50
High dose	60 mg	3	0	50

2.5 Outcomes

The primary outcome was the change in fatigue from baseline to week 12, measured with the Fatigue Severity Index, a 30-point self-report scale on which higher scores mean more fatigue. The index has been validated in adult outpatients and has good internal consistency and test-retest reliability. Participants completed it at baseline and at weeks 4, 8 and 12, without access to their earlier answers.

Secondary outcomes were serum ferritin, transferrin saturation and haemoglobin at baseline and week 12, measured in fasting morning samples in the hospital laboratory; adherence; and adverse events. Adverse events were recorded at every visit with a structured checklist of gastrointestinal symptoms followed by an open question.

2.6 Sample size

The trial was designed to estimate the dose-response slope with useful precision rather than to test a difference between two arms. Based on the variability of fatigue change in earlier trials, 50 participants per arm were expected to give a confidence interval for the slope narrow enough to distinguish a clinically relevant effect from none, allowing for up to 10% loss to follow-up.

2.7 Statistical analysis

Analyses followed the intention-to-treat principle and included all randomised participants. The change in the fatigue score was analysed with a linear mixed model with participant as a random effect, daily dose as a continuous fixed effect, visit as a categorical fixed effect, and baseline fatigue score and sex as covariates. The dose-response slope was expressed as the reduction in fatigue score per 10 mg of daily iron, with a 95% confidence interval. Missing outcomes were handled by the mixed model under the assumption that data were missing at random; a sensitivity analysis used multiple imputation. Departure from linearity was examined by adding a quadratic term for dose. Secondary outcomes were compared between arms with analysis of covariance adjusted for the baseline value. All analyses were done with standard statistical software and a two-sided significance level of 5%.

3. Results

3.1 Participants

Of 412 adults screened, 150 were eligible and were randomised, 50 to each arm. Recruitment took place over 14 months. The most common reasons for exclusion were a ferritin above the threshold, anaemia and a raised C-reactive protein. In total, 144 participants completed the 12-week visit; the six who did not were lost to follow-up or withdrew for reasons unrelated to the study capsules in four cases and because of gastrointestinal complaints in two. The three arms were similar at baseline in age, sex, body mass index, iron status and fatigue score (Table 2).

Figure 1. Flow of participants through the trial: screening, randomisation to 20, 40 or 60 mg of elemental iron daily, follow-up and analysis.

Table 2. Baseline characteristics of the randomised participants by arm (mean or number)

Characteristic	20 mg (n = 50)	40 mg (n = 50)	60 mg (n = 50)
Age, years	31.8	32.6	31.2
Women, n	41	42	41
Body mass index, kg/m²	22.9	23.4	23.1
Serum ferritin, µg/L	15.2	14.7	15.0
Transferrin saturation, %	17.9	18.3	17.6
Haemoglobin, g/dL	13.1	13.2	13.0
Fatigue Severity Index (0–30)	18.4	18.9	18.6

3.2 Iron status

Serum ferritin rose in all three arms over the 12 weeks, and the rise was larger with higher doses. Transferrin saturation followed the same pattern. Haemoglobin, which was normal at baseline, changed little in any arm. The proportion of participants whose ferritin reached at least 30 µg/L by week 12 increased from the low-dose arm to the high-dose arm (Table 4). The increase in ferritin per extra 10 mg of iron was smaller between 40 and 60 mg than between 20 and 40 mg, which is consistent with reduced fractional absorption at higher doses.

Figure 2. Mean serum ferritin at baseline and week 12 in each arm. Error bars show 95% confidence intervals.

3.3 Fatigue

Fatigue scores fell in all three arms during the trial, with most of the fall occurring in the first eight weeks. The reduction at week 12 was larger in the higher-dose arms. In the linear mixed model, the reduction in fatigue increased steadily with the daily dose of iron. Table 3 gives the dose-response slope: 0.28 points per 10 mg (95% CI 0.15 to 0.41). Adding a quadratic term for dose did not improve the fit of the model, so there was no evidence that the effect levelled off within the range studied. The multiple-imputation sensitivity analysis gave a similar estimate. The slope was similar in women and men, although the number of men was small.

Table 3. Dose-response of the change in Fatigue Severity Index from baseline to week 12 (linear mixed model, intention to treat, n = 150)

Estimate	Value	95% CI
Dose-response slope, points per 10 mg of daily iron	0.28	0.15 to 0.41

Figure 3. Mean Fatigue Severity Index at baseline and at weeks 4, 8 and 12 in each arm. Lower scores mean less fatigue.

3.4 Adherence and adverse events

Adherence by capsule count was high in all arms but slightly lower in the 60 mg arm. Gastrointestinal adverse events were reported by more participants as the dose increased; the most frequent were nausea, constipation and dark stools. No serious adverse events occurred, and two participants in the 60 mg arm stopped taking the capsules because of abdominal discomfort.

Table 4. Iron status at week 12, adherence and adverse events by arm

Outcome	20 mg	40 mg	60 mg
Ferritin ≥ 30 µg/L at week 12, %	38	56	64
Mean adherence, % of capsules	93	91	86
Any gastrointestinal adverse event, n	9	14	22
Nausea, n	4	6	11
Constipation, n	3	5	8
Stopped capsules because of adverse events, n	0	0	2

4. Discussion

In this randomised trial in non-anaemic adults with iron deficiency, the reduction in fatigue over 12 weeks grew with the daily dose of oral iron across the range of 20 to 60 mg, with no sign that the benefit levelled off. At the same time, gastrointestinal complaints became more frequent and adherence slightly lower as the dose increased.

Our findings agree with earlier placebo-controlled trials that found less fatigue after iron in women with low ferritin, and extend them by describing how the effect depends on dose. The pattern in ferritin, with smaller gains per milligram at the highest dose, fits absorption studies showing that larger doses raise hepcidin and lower the fraction absorbed. It is notable that fatigue continued to improve with dose even though the gain in ferritin per milligram became smaller, which suggests that the symptom may respond to the amount of iron absorbed rather than to the stores measured by ferritin.

The trial has several strengths. Allocation was concealed and participants and assessors were blinded, the capsules were identical in all arms, loss to follow-up was low, and the dose was modelled as a continuous variable, which uses all three arms to estimate a single slope. Fatigue was measured with a validated scale at four time points.

There are also limitations. The trial had no placebo arm, so the slope describes the difference between doses and not the effect of iron against none; part of the fall in fatigue in every arm may reflect regression to the mean or expectation. Most participants were young women, and the results may not apply to older adults, men or people with anaemia. Twelve weeks may be too short to see the full effect of the lower doses, which might catch up with longer treatment. Fatigue is self-reported and subjective, although the blinding reduces the risk that expectation differed between arms. Finally, we studied a single iron salt taken daily, and alternate-day dosing or other formulations may behave differently.

For practice, the results suggest that higher daily doses relieve fatigue more within this range, but

the gain must be weighed against more frequent side effects, which may reduce adherence outside a trial. Future trials should include a placebo arm, test alternate-day schedules and follow participants for longer.

5. Conclusions

In iron-deficient, non-anaemic adults, the reduction in fatigue after 12 weeks of oral iron increased with the daily dose between 20 and 60 mg, while gastrointestinal adverse events also became more common. Dose choice should balance both effects.

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